

Which data structure is used during recursion

- A. merge sort
- B. stack
- C. list
- D. linear search

ANSWER: B

Postfix conversion uses_____ concepts?

- A. List
- B. Graph
- C. Tree
- D. Stack

ANSWER: D

Where can we perform insertion on stack?

- A. Front
- B. Rear
- C. Middle
- D. Top

ANSWER: D

If front pointer floating in a queue, then it is termed as

- A. Queue
- B. Priority queue
- C. Floating front queue
- D. Circular queue

ANSWER: C

What are the disadvantages of arrays?

- A. Index value of an array can be negative
- B. Elements are sequentially accessed
- C. Data structure like queue or stack cannot be implemented
- D. There are chances of wastage of memory space if elements inserted in an array are lesser than the allocated size

ANSWER: D

What is the value of the postfix expression 6 3 2 4 + - *?

- A. 74
- B. -18
- C. 22
- D. 40

ANSWER: B

Which of the following statement(s) about stack data structure is/are NOT correct?

- A. Top of the Stack always contain the new node
- B. Stack is the FIFO data structure
- C. Null link is present in the last node at the bottom of the stack
- D. Linked List are used for implementing Stacks

ANSWER: B

The prefix form of $A - B / (C * D ^ E)$ is?

- A. $-A/B * C ^ D E$
- B. $-A/BC * ^ D E$

- C. $-ABCD*^{\wedge}DE$
- D. $-/*^{\wedge}ACBDE$

ANSWER: A

The optimal data structure used to solve Tower of Hanoi is _____

- A. Tree
- B. Heap
- C. Priority queue
- D. Stack

ANSWER: D

Which is the most appropriate data structure for reversing a word?

- A. stack
- B. queue
- C. graph
- D. tree

ANSWER: A

How can we describe an array in the best possible way?

- A. The Array shows a hierarchical structure.
- B. Arrays are immutable.
- C. Container that stores the elements of similar types
- D. The Array is not a data structure

ANSWER: C

How can we initialise an array in C language?

- A. `int arr[2]=(10, 20)`
- B. `int arr(2)={10, 20}`
- C. `int arr[2] = {10, 20}`
- D. `int arr(2) = (10, 20)`

ANSWER: C

If the size of the stack is 10 and we try to add the 11th element in the stack then the condition is known as____

- A. Underflow
- B. Garbage collection
- C. Overflow
- D. None of the option

ANSWER: C

What is the outcome of the prefix expression +, -, *, 3, 2, /, 8, 4, 1?

- A. 12
- B. 11
- C. 5
- D. 4

ANSWER: C

Which one of the following is the overflow condition if a circular queue is implemented using array having size MAX?

- A. `rear= MAX-1`
- B. `rear=MAX`
- C. `front=(rear+1) mod max`
- D. None of the option

ANSWER: C

What is the time complexity of inserting an element in a stack implemented using an array, if the stack is already full?

- A. $O(\log n)$
- B. $O(2n)$
- C. $O(1)$
- D. $O(n)$

ANSWER: D

Which data structure is typically used to implement Undo functionality in text editors?

- A. stack
- B. queue
- C. graph
- D. tree

ANSWER: A

Identify the data structure which allows deletions at both ends of the list but insertion at only one end

- A. Input-restricted deque
- B. Output-restricted deque
- C. Priority queues
- D. None of above

ANSWER: A

What is not true about insertion sort?

- A. Exhibits the worst case performance when the initial array is sorted in reverse order
- B. Worst case and average case performance is $O(n^2)$
- C. Can be compared to the way a card player arranges his card from a card deck
- D. None of the above

ANSWER: D

If queue is implemented using arrays, what would be the worst run time complexity of queue and dequeue operations?

- A) $O(n), O(1)$
- B) $O(1), O(1)$
- C) $O(1), O(n)$
- D) $O(n), O(1)$

ANSWER: B

An item that is read as input can be either pushed to a stack and later popped and printed, or printed directly. Which of the following will be the output if the input is the sequence of items - 1,2,3,4,5?

- A) 3,4,5,1,2
- B) 3,4,5,2,1
- C) 1,5,2,3,4
- D) 5,4,3,1,2

ANSWER: B

If the characters 'D' 'C' 'B' 'A' are placed in a queue (in that

order), and then removed one at a time , in what order will they be removed?

- A) DCAB
- B) ABCD
- C) ABDC
- D) DCBA

ANSWER: D

If the sequence of operations:

push(1),push(2),pop,push(1),push(2),pop,pop,pop,push(2),pop, are performed on a stack, the sequence of popped out values are

- A) 2,2,1,1,2
- B) 2,2,1,2,2
- C) 2,1,2,2,1
- D) 2,1,2,2,2

ANSWER: A

In a circular queue the value of r will be

- A) $r=r+1$
- B) $r=(r+1)\%[\text{queue_size}-1]$
- C) $r=(r+1)\%\text{queue_size}$
- D) $r=(r-1)\%\text{queue_size}$

ANSWER: C

In a stack the command to access nth element from top of the stack s will be

- A) $S[\text{top}+n]$
- B) $S[\text{top}-n-1]$
- C) None of the option

ANSWER: C

In an empty queue, rear and front can be initialized as

- A) $\text{rear}=\text{front}=-1$
- B) $\text{rear}=\text{front}=1$
- C) $\text{rear}=0, \text{front}=-1$
- D) $\text{rear}=-1, \text{front}=0$

ANSWER: A

Queues serve major role in

- A) Simulation of recursion
- B) Simulation of arbitrary linked list
- C) Simulation of limited resource allocation
- D) All of the mentioned

ANSWER: C

Stack A has the entries a,b,c (with a on top). Stack B is empty. An entry popped out of stack A is pushed into stack B. An entry popped out of stack B can only be printed. In this arrangement, which of the following permutations of a,b,c is not possible?

- A) b a c
- B) b c a
- C) c a b
- D) a b c

ANSWER: C

Suppose you are given an array $A[1, \dots, n]$ and a procedure $\text{reverse}(A, i, j)$ which reverse the order of elements in between positions i and j . What does the sequence $\text{reverse}(A, 1, i); \text{reverse}(A, i+1, n); \text{reverse}(A, 1, n)$ do where $1 \leq i \leq n$;

- A) Rotates A left by i positions
- B) Leaves A unchanged
- C) Reverses all elements of A
- D) None

ANSWER: A

The five items: A,B,C,D and E are pushed in stack, one after the other starting from A. The stack is popped four items and each element is inserted in a queue. Then two elements are deleted from the queue and pushed back on the stack . Now one item is popped from the stack. The popped item is.

- A) A
- B) B
- C) C
- D) D

ANSWER: D

The initial configuration of a queue is a,b,c,d, ('a' is in the front end). To get the configuration d,c,b,a, one needs a minimum of

- A) 2 deletions and 3 additions
- B) 3 deletions and 2 additions
- C) 3 deletions and 3 additions
- D) 3 deletions and 4 additions

ANSWER: C

The number of swappings needed to sort the numbers 8,22,7,9,31,5,13 in ascending order using bubble sort is

- A) 11
- B) 12
- C) 13
- D) 10

ANSWER: D

The postfix expression for the infix expression $A+B*(C+D)/F+D*E$ is

- A) $AB+CD+*F/D+E*$
- B) $ABCD+*F/+DE*++$
- C) $A*B+CD/F*DE++$
- D) $A+*BCD/F*DE++$

ANSWER: B

The postfix form of $A-B/(C*D\$E)$ is

- A) $ABCDE\$*/-$
- B) $AB/C*DE\$$
- C) $ABCDE\$-/*$
- D) $ABCDE/-*\$$

ANSWER: A

The result of evaluating the postfix expression $10\ 5\ +\ 60\ 6\ /\ * \ 8\ -$ is

- A) 284
- B) 213
- C) 142
- D) 71

ANSWER: C

The result of evaluating the prefix expression $*/b+-dacd$, where $a=3$, $b=6$, $c=1$, $d=5$ is

- A) 0
- B) 5
- C) 10
- D) 15

ANSWER: C

The sequence of operations is performed on a stack
PUSH(10), PUSH(20), POP, PUSH(10), PUSH(20), POP, POP, POP, PUSH(20), POP.
The sequence of values popped out is

- A) 20, 10, 20, 10, 20
- B) 20, 20, 10, 10, 20
- C) 10, 20, 20, 10, 20
- D) 20, 20, 10, 20, 10

ANSWER: B

What can we say about the array representation of a circular queue when it contains only one element?

- A) Front equal Rear equal Null
- B) Front equal Rear not equal Null
- C) Front equal Rear equal -1
- D) None

ANSWER: B